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CO-2 Assessment Tool-2

1. Ans @ Data Mining & KDD

The process of discovering hidden, useful patterns and knowledge from large datasets.

Two Basic Tasks: 1. Classification, 2. Clustering

KDD Process Steps:

1. Selection: Choosing the relevant data.
2. Preprocessing: Cleaning and combining data.
3. Transformation: Formatting data for mining.
4. Data Mining: Applying algorithms to extract patterns.
5. Evaluation: Identifying truly useful patterns.
6. Presentation: Visualizing the extracted knowledge.

Role of Preprocessing: Real data is messy. Preprocessing cleans and structures it, which is essential to accurate, reliable, and fast data mining results.

⑥ Data Cleaning & Quality:-

Detecting and fixing corrupt, incomplete or inaccurate records in a dataset.

Two Quality Issues: 1. Missing Values, 2. Noise

Handling Missing Values:

- * Ignore the row: Delete the entire record if crucial info is missing.
- * Global Constant: Fill blanks with a label like "Unknown".
- * Mean/Median: Replace the blank with the average value of that attribute.

Handling Noisy Data:

- * Binning: Sort data into buckets and smooth them by replacing values with the bin's average.
- * Regression: Fit data to a trend line; data points far from the line are treated as noise.

2. Ans:

① Data Integration & Challenges

Combining data from multiple different sources into one unified data store, like a data warehouse.

Two Challenges: - 1. Entity Identification, 2. Data Value Conflicts.

Handling Redundancy & Inconsistency:

- * Numerical Data: Use Correlation Analysis to detect and remove highly correlated attributes.

- * Categorical Data: Use a Chi-Square test to find redundant nominal attributes
- * Metadata & Transformation: Use metadata to map mismatched names, and apply data transformations to standardize units and formats.

⑥ Covariance Analysis & Formulas

A statistical measure that determines the directional relationship between two numeric variables

- * Positive Covariance: Both variables increase or decrease together.
- * Negative Covariance: As one increases, the other decreases
- * Zero Covariance: The variables are independent.

Two formulas:

1. Standard formula:

$$\text{Cor}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

2. Simplified/Computational formula:

$$\text{Cor}(X, Y) = \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right) - \bar{x} \bar{y}$$

3. Ans: (a) Stock Price Movement:

Given, Stock A(x) and Stock B(y)

values $n=5$

Means

Average of A($\bar{x}$) = 4

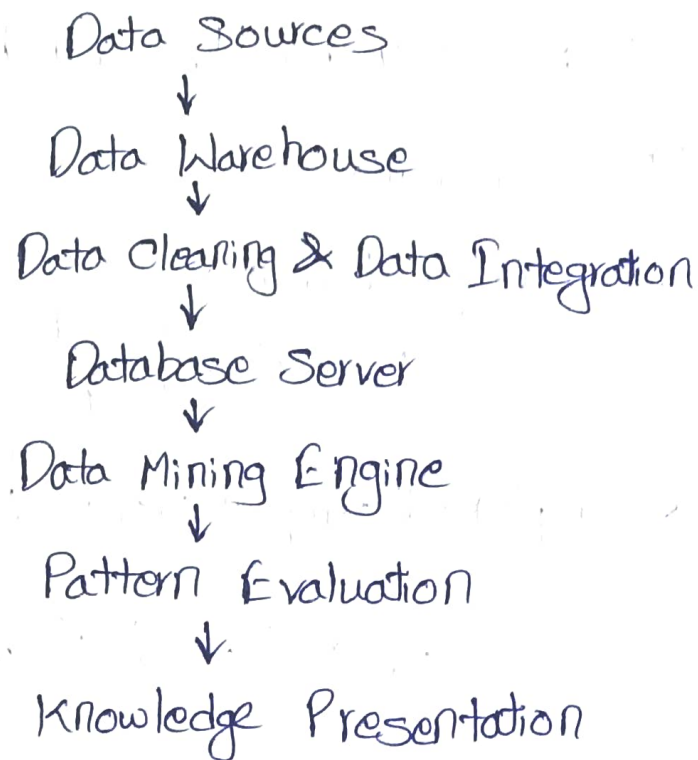
Average of B($\bar{y}$) = 9.6

$$\text{Cov}(X, Y) = (\text{Average of } X \cdot Y) - (\bar{x} \cdot \bar{y})$$

$$\text{Cov}(X, Y) = \left(\frac{215}{5}\right) - (4 \times 9.6) = 42.4 - 38.4 = 4$$

Conclusion:- Because the covariance is positive (+4), the stocks are not independent. Their prices will rise and fall together.

(b) Data Mining System Architecture:



Core Components:

1. Repositories
2. Knowledge Base
3. Data Mining Engine
4. Pattern Evaluation
5. GUI

Ans: (a) Covariance

Given $X = \{4, 5, 6, 7, 9\}$ $Y = \{8, 3, 7, 5, 6\}$ $n = 5$

$$\text{Means } \bar{x} = \frac{4+5+6+7+9}{5} = \frac{31}{5} = 6.2 \quad \bar{y} = \frac{8+3+7+5+6}{5} = \frac{29}{5} = 5.8$$

Sum of products

$$(4 \times 8) + (5 \times 3) + (6 \times 7) + (7 \times 5) + (9 \times 6) = 32 + 15 + 42 + 35 + 54 \\ = 178$$

Covariance:

$$\text{Cov}(X, Y) = \left(\frac{\sum xy}{n} \right) - (\bar{x} \cdot \bar{y}) = \left(\frac{178}{5} \right) - (6.2 \times 5.8) \\ = 35.6 - 35.96 = -0.36$$

Conclusion: The negative covariance indicates a slight inverse relationship between X and Y.

(b) Data Pre-processing Steps & Pipeline

1. Major Steps in Data Pre-Processing

- * Data Cleaning
- * Data Integration
- * Data Reduction

* Data Transformation

2. Systematic Pipeline for the Given Dataset

* Resolve Inconsistent Formats

* Handle outliers

* Fill Missing values

5. Ans: (a) Equal-Frequency Binning

Given 27 ages

Divide into equal Bins: $27/3 = 9$

Bin 1:- 13, 15, 16, 16, 19, 20, 20, 20, 21, 22 [mean = 18]

Bin 2:- 22, 25, 25, 25, 25, 30, 33, 33, 35 [mean = 28.11]

Bin 3:- 35, 35, 35, 36, 40, 45, 46, 52, 70 [mean = 43.78]

(b) Data Discretization & Binning Comparison

Converting continuous data into finite discrete intervals. Importance: Required by many algorithms, reduces noise, and speeds up data processing.

* Equal Width Binning $\rightarrow$ Poor at handling outliers

* Equal Frequency Binning $\rightarrow$ Excellent for handling outliers.